IGCSE 0607 Practice Questions — E1.16

Exponential growth and decay

Total Questions: 12 — **Total Marks:** 63 Marks

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1 A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

(a) **(a)** Work out the value of p.

$p =$ ..... **[3]**

(b) Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

2 A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

3 A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

4 The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

(a) Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

(b) Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

5 The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

(a) **(a)** Calculate the value of r, correct to 3 significant figures.

$r =$ ..... **[3]**

(b) Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

6 A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

(a) Calculate the value of x.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

7 Consider the depreciation of two different vehicles over time.

(a) A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year. Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

(b) A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

8 The mass, M grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Find the initial mass of the substance.

..... grams **[2]**

(c) Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

9 The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

(a) Calculate the percentage increase in the population per week.

..... % **[4]**

(b) Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

(c) Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

10 A new car is bought for \$26000. The value of the car depreciates exponentially at a rate of 15% each year.

Work out the value of the car after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[3]**

11 A population of bacteria in a dish grows exponentially.

(a) Initially, there are 1500 bacteria in the dish.
The population increases by 8% every hour.
Work out the population of bacteria after 12 hours.
Give your answer correct to the nearest whole number.

..... **[2]**

(b) Work out the least number of complete hours it will take for the initial population of 1500 bacteria to be greater than 3000.

..... hours **[3]**

12 **(a)** The value of a painting increases exponentially by $r\%$ each year.
After 2 years, the total percentage increase in the value of the painting is 44%.
Work out the value of r.

$r =$ ..... **[3]**

(b) The value of a rare stamp increases exponentially.
Its value doubles every 15 years.
The current value of the stamp is \$80.

Work out the value of the stamp after 45 years.

% ..... **[2]**